

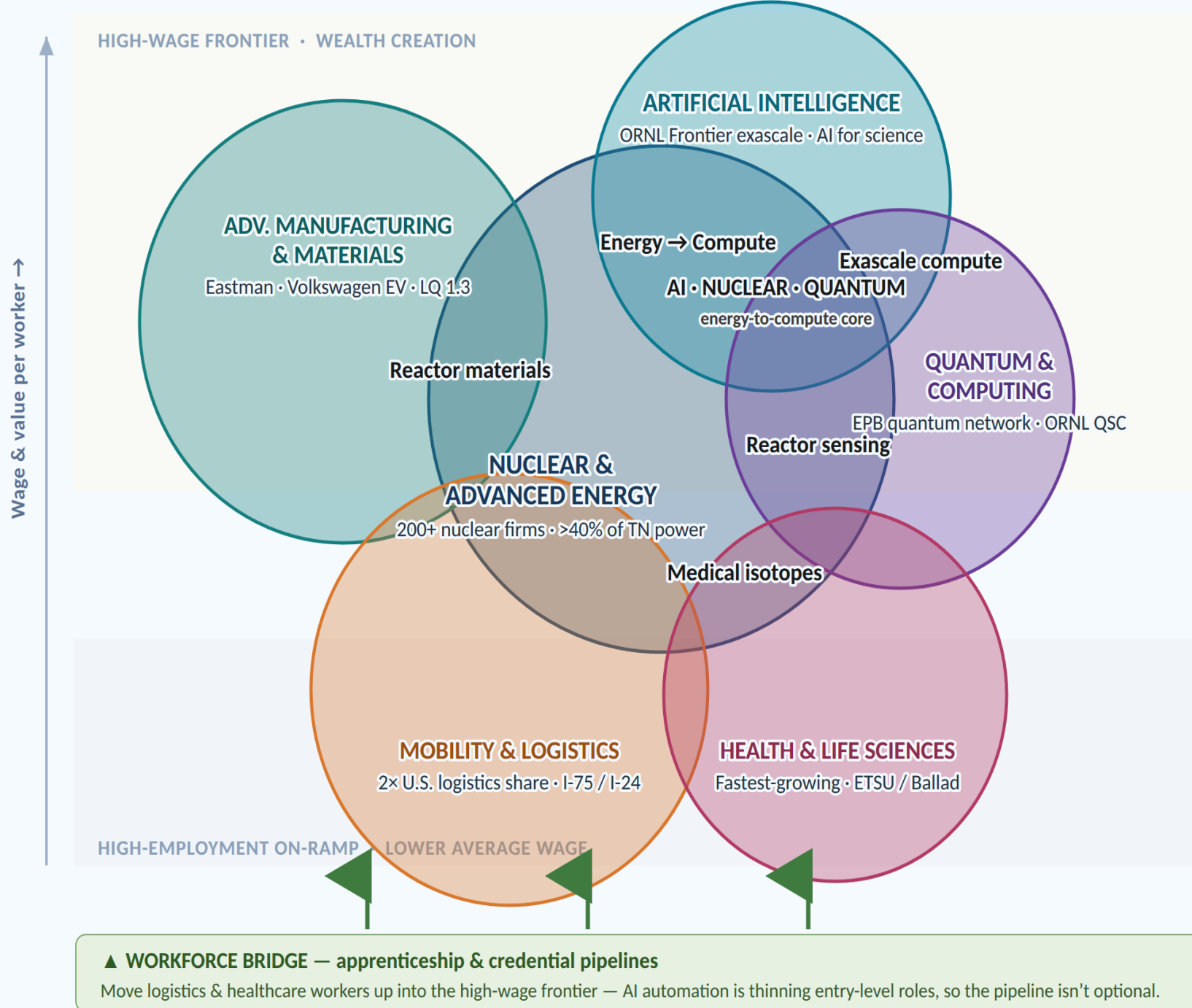


East Tennessee: Foundation to Frontier

A Regional Competitiveness Strategy — Tourism Cluster · Nuclear Cluster

New Nuclear Cluster

Foundation → Keystone → Frontier and where they converge



HOW TO READ THIS MAP

- Bubble size ≈ employment scale
- Height ≈ wage / value per worker
- Overlap = shared traded work

CLUSTERS

- Nuclear & Advanced Energy
- Artificial Intelligence — cross-cutting
- Quantum & Computing
- Adv. Manufacturing & Materials
- Mobility & Logistics
- Health & Life Sciences

CONVERGENCE ZONES

- AI • Nuclear • Quantum core
- Energy-to-Compute (Nuclear × AI)
- Exascale compute (AI × Quantum)
- Medical isotopes (Nuclear × Health)

TWO-TIER STRATEGY

- Capture** the frontier core — AI • Nuclear • Quantum (energy-to-compute).
- Build** the workforce bridge so on-ramp clusters (Logistics, Health) feed it.
- Diversify** via Health — Mo-99 ties to Nuclear.

Tourism is the fiscal foundation • New Nuclear the keystone • AI & Quantum the frontier — linked by the energy-to-compute core.

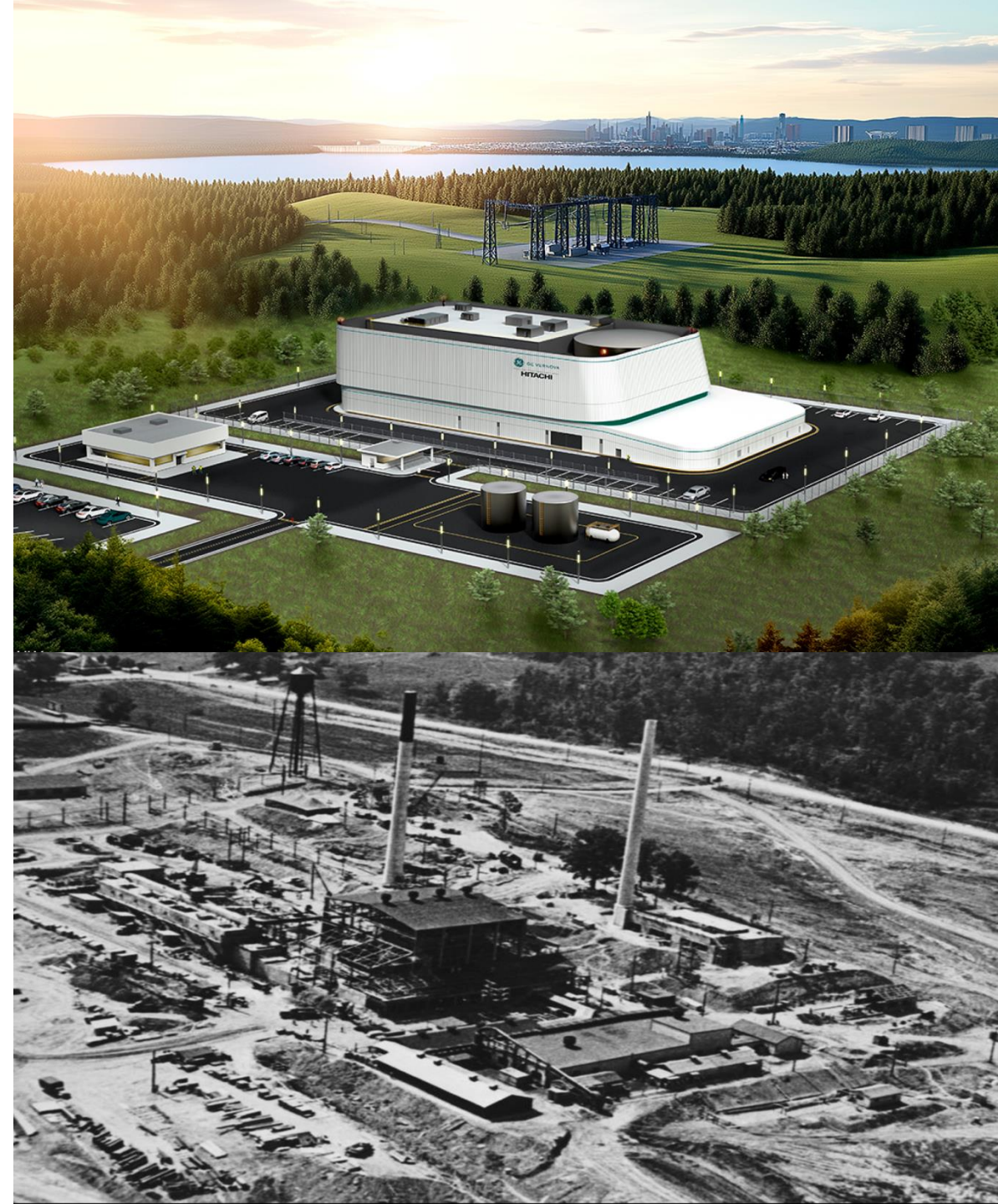
Frontier heritage, modern commercialization

East Tennessee is America's nuclear commercialization corridor — where next-generation nuclear moves from discovery to deployment — and East Tennessee's most accessible mountains-city-water destination.

Unique strengths

- Advanced energy innovation + real-world power (TVA)
- World-class R&D and talent — ORNL, University of Tennessee
- Deep supplier capability + live deployment sites
- Iconic visitor economy and exceptional quality of place

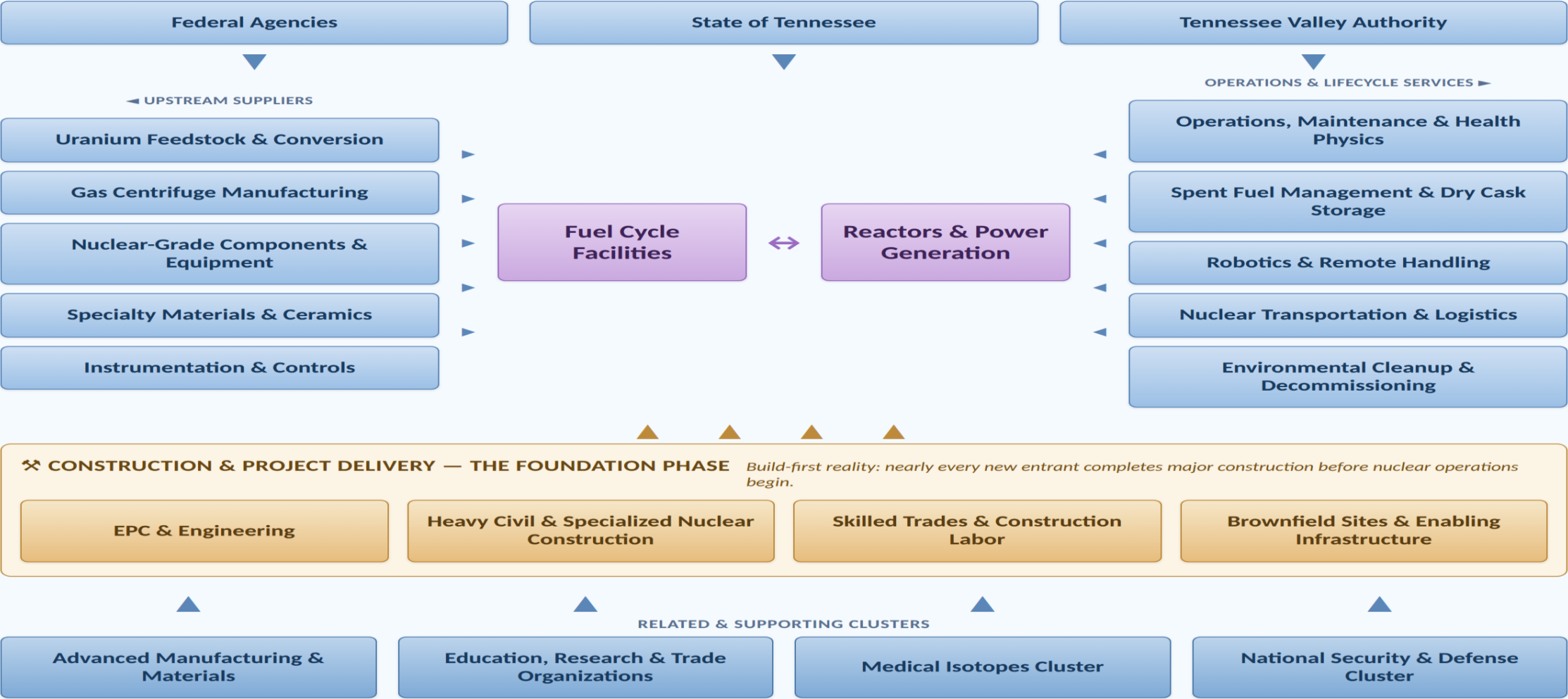
“Companies come here to build, test, validate, commercialize, and scale.”



The East Tennessee Nuclear Cluster

The nuclear fuel cycle, the upstream supply chain, and the construction phase that precedes every new entrant — across the Oak Ridge–Knoxville–Tri-Cities corridor

GOVERNMENT & INSTITUTIONAL FRAMEWORK



Structure: Modeled on M. Porter & C. Bond, "The California Wine Cluster," Harvard Business School. Purple = core value chain; blue = supplier inputs, lifecycle services, and adjacent clusters; amber = the construction/project-delivery phase that precedes nuclear operations for new entrants.
Sources: DOE / NNSA / DOE-EM, U.S. NRC, TNECD, East Tennessee Economic Council Nuclear Industry Hub, ORNL, Bechtel, and company announcements, 2025–2026.

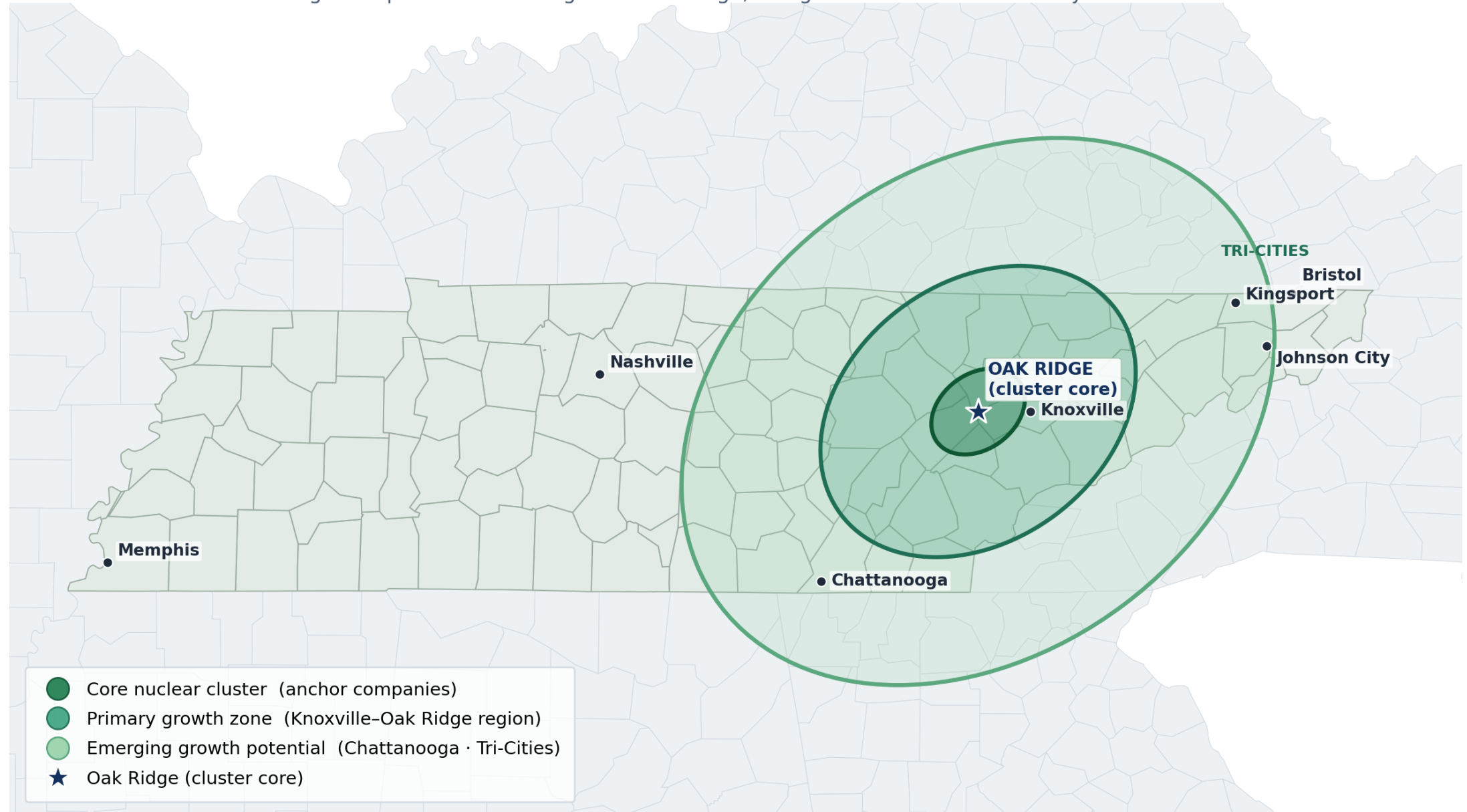
Build-first reality: nearly every new entrant constructs before it goes nuclear.

Tennessee Nuclear Cluster – Oak Ridge Epicenter

Collaboration, partnerships, and impacts reach beyond the Knoxville-Oak Ridge region

Tennessee Nuclear Cluster – Growth from the Oak Ridge Core

Concentric zones of cluster growth potential radiating from Oak Ridge, along the East Tennessee valley corridor



County boundaries: U.S. Census (2018). Ellipses are conceptual zones of cluster growth potential, major axis aligned to TN's eastern edge (~37°).

Make East Tennessee the new nuclear commercialization hub

Establish East Tennessee as the leading integrated hub for nuclear commercialization, supply-chain development, and real-world deployment.

Why this priority

- Unique concentration across R&D, utility operations, manufacturing, workforce & deployment
- Compute demand for AI & quantum is growing exponentially
- Advanced nuclear is moving from research to commercialization — a time-limited window
- Value accrues to places that control commercialization, manufacturing & deployment
- Creates high-wage jobs, export industries, talent attraction & cross-sector spillovers



Confident, deployment-first ambition — “looking for a bigger target.”

Nuclear Commercialization Execution — 12-Month Plan

Regional effort with statewide partnership and impact



Nuclear Commercialization

1. Launch a Nuclear Supply Chain Working Group
2. Establish a Nuclear Pilot & Deployment Program
3. Continue executing targeted company recruitment
4. Launch Founder & Supplier Development to enable the supply chain

Nuclear Commercialization - 90-Day Activation & First-Year Targets

Days 1–30

Convene leadership; adopt the regional ambition

Deliverable: signed charter

Days 31–60

Draft the regional cluster map

Inventory, capability gaps, recruitment targets

Days 61–90

Launch first visible initiatives

Pilot · venture fund · recruitment mission · supplier initiative

First-year targets

10+

company recruitment opportunities

3+

pilots launched

15+

new nuclear firms engaged

\$25M+

private capital committed

250+

students & workers connected

Frontier heritage, traditional tourism, modern amenities

East Tennessee's natural resources provide a variety of opportunities from traditional offerings to adventurous outings for all.

Unique strengths

- A day's drive from 75% of the Nation's population
- Great Smoky Mountain's National Park is the most visited park in the Nation
- A hub for adventure tourism (rafting, hiking, biking)
- 2 of the top 4 counties for visitation are in East Tennessee (Knox, Sevier – with all 16 counties in the top third for the state of Tennessee)

"The Smokies always call me home".



East Tennessee Tourism, by the Numbers

The region's visitor economy across the East Tennessee grand division — 2024



\$9.4B+

combined visitor spending — East TN



12.2M

Great Smoky Mountains visits — the most-visited U.S. national park



\$3.93B

Sevier County (Gatlinburg · Pigeon Forge) — 3rd of 95 TN counties



3 of 5

of Tennessee's top visitor-spending counties are in East Tennessee



\$2.8B

the Smokies' benefit to gateway communities — highest of any national park



3.14M

Dollywood guests — a top-20 North American theme park

Within Tennessee's record \$31.7B visitor economy (147M visits, \$3.3B state & local tax). Sources: TDTD / Tourism Economics 2024; NPS; TEA.

When we succeed — East Tennessee, 2036

- The nation's new nuclear commercialization capital — build, test & scale here
- An energy-to-compute flywheel powering AI & quantum on clean, firm nuclear power
- A visitor economy that brings benefits and keeps talent throughout the 16-county region and beyond
- High-wage jobs, export industries, and a magnet for people and capital



**Partnerships are built before
you need them.**



Thank you

East Tennessee — Foundation to Frontier

Knoxville—Oak Ridge Region · Questions & discussion

